

Integrating Process Safety in Engineering Project - Best Practices

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Focus Area



Process Safety and its integration



Process Incidents



Types of Projects and its Life-cycle



Why is integration required?



Nayara Energy's approach

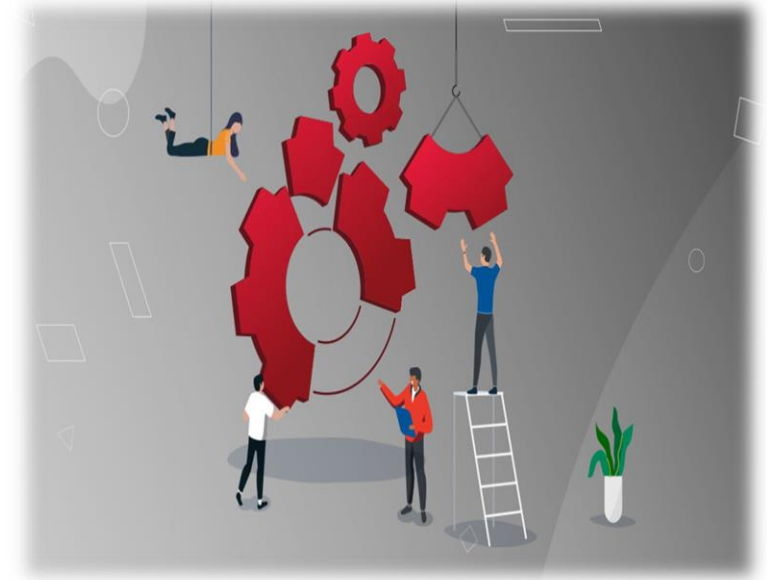


Conclusion

Process Safety:

Process Safety is a systematic way (identify, evaluate, and mitigate the process risk) to manage the Process systems and assets to ensure that it will operate as per the design intent, without impacting Human, Environment and Assets.

Process safety helps to manage highly hazardous materials and energy to prevent catastrophic events.



Process Safety integration:

Process Safety incidents often originate from poor design decisions made early in a project. Once a facility is built, correcting these issues becomes expensive or nearly impossible.

Integrating process safety from the beginning ensures that risks are eliminated or minimized **by design**—the most effective and sustainable approach.



Process Safety Incidents:

Process Safety incident is an unplanned or uncontrolled release of any material, energy, or substance (including toxic, flammable, reactive, or high-pressure materials) from a process that could cause harm to **People, Environment, Assets and Reputation** of the organisation.

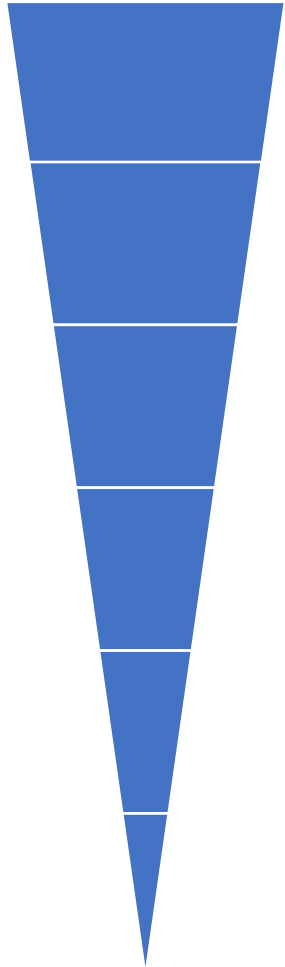


Engineering Projects:

- Green field, Brown field & Debottleneck
- Indoor & Outdoor, Onshore & Offshore
- Major Modifications (covered by MOC)
- Loading and offloading systems
- Control systems
- Utility systems
- Buildings
- Pipelines, etc.



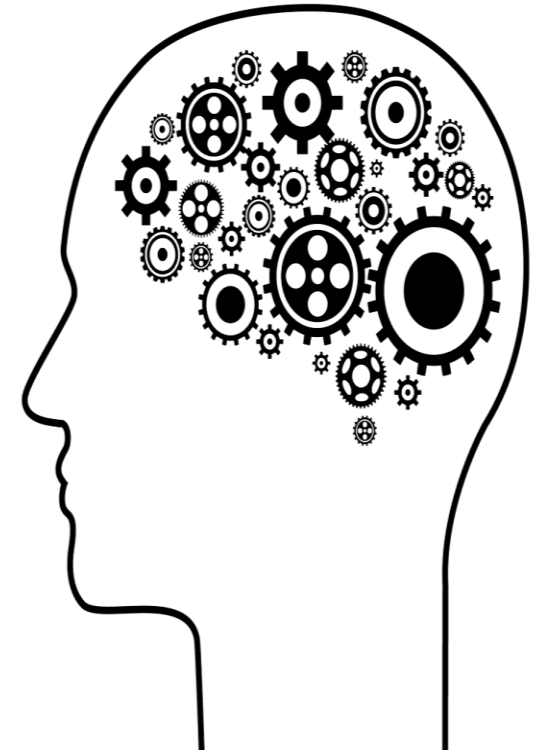
Project Life Cycle:



- Concept or Feasibility Stage
- Basic Design Engineering Package
- Front End Engineering Design
- Detailed Engineering
- Construction
- Pre-commissioning & Commissioning
- Handover to Operations

Integration Objective:

- Inherently safer design (ISD)
- Identification of high risks
- Statutory and Regulatory compliance
- Late stage design changes
- Project reliability, operability and efficiency
- Reduce the like hood and severity of catastrophic releases
- Reduce risk to the People, Environment, Assets and Reputation

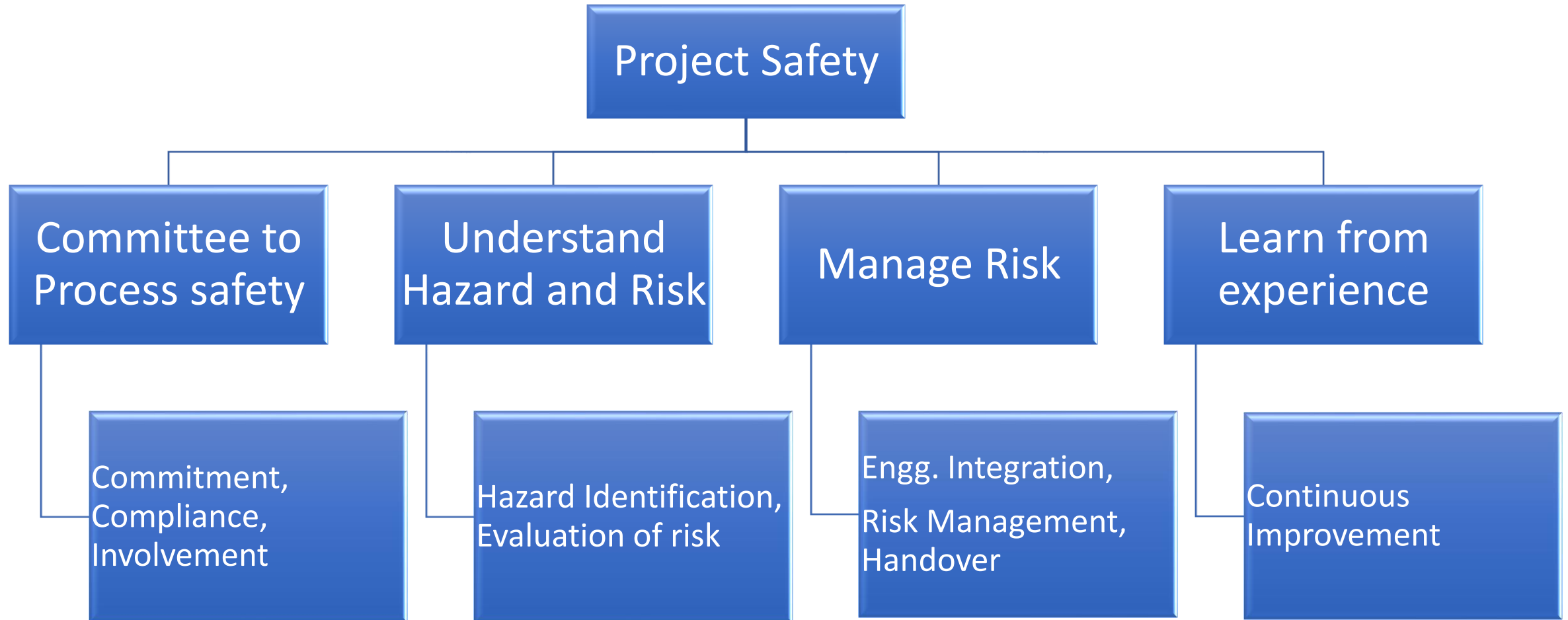


How to integrate:

- Stage of the Project
- Integrate with engineering
- Integrate with the compliances
- Integrate with Contractors
- Integration the Leadership
- Manage the Changes
- Handover
- Continuous improvement



How Nayara's integration is ?



Commitment:

- ✓ Ownership and Key responsibilities
- ✓ Competency and culture
- ✓ Safety prioritized over cost and schedule
- ✓ Contractors engagement & commitment



Compliances:

- ✓ Develop compliance register
- ✓ Monitoring & Audit
- ✓ Incorporate requirements in Design Basis
- ✓ Early plan and timely approvals



Involvement :

- ✓ Employee involvement, commitment and expectations
- ✓ Contractors involvement and alignment in HSE Plans
- ✓ Multidiscipline team involvement in safety studies
- ✓ Multilevel team involvement in critical activities

Stage of the Project:

- ✓ Include early process safety experts from project inception
- ✓ Inherently Safer Design (ISD) approach
- ✓ Develop risk acceptance criteria's and risk matrix
- ✓ Conduct Process Hazard identification and evaluation
- ✓ Develop the safety design and philosophy documents
- ✓ Develop and maintain Safety deliverables



Integrate with Engineering:

- ✓ Process design
- ✓ Mechanical design
- ✓ Electrical
- ✓ Instrumentation
- ✓ Civil & layout design
- ✓ Safety system designs
- ✓ Avoid retrofitting safety at later stages



Risk Management

- ✓ Identification and mitigation of construction risks
- ✓ Identification and mitigation of Precom and Comm. risk
- ✓ Verification of all installation by SME's before PSSR
- ✓ Compliance of MOC process during projects
- ✓ Compliance of ERP
- ✓ Pre-Startup Safety Review (PSSR) compliance
- ✓ Review and verification of PSI documents
- ✓ Develop or follow the SOP and SWP



Handover:

- ✓ Maintain updated PSI's and safety studies
- ✓ Handover complete safety documentation
- ✓ Ensure operations understand design intent
- ✓ Preserve lessons learned for future projects

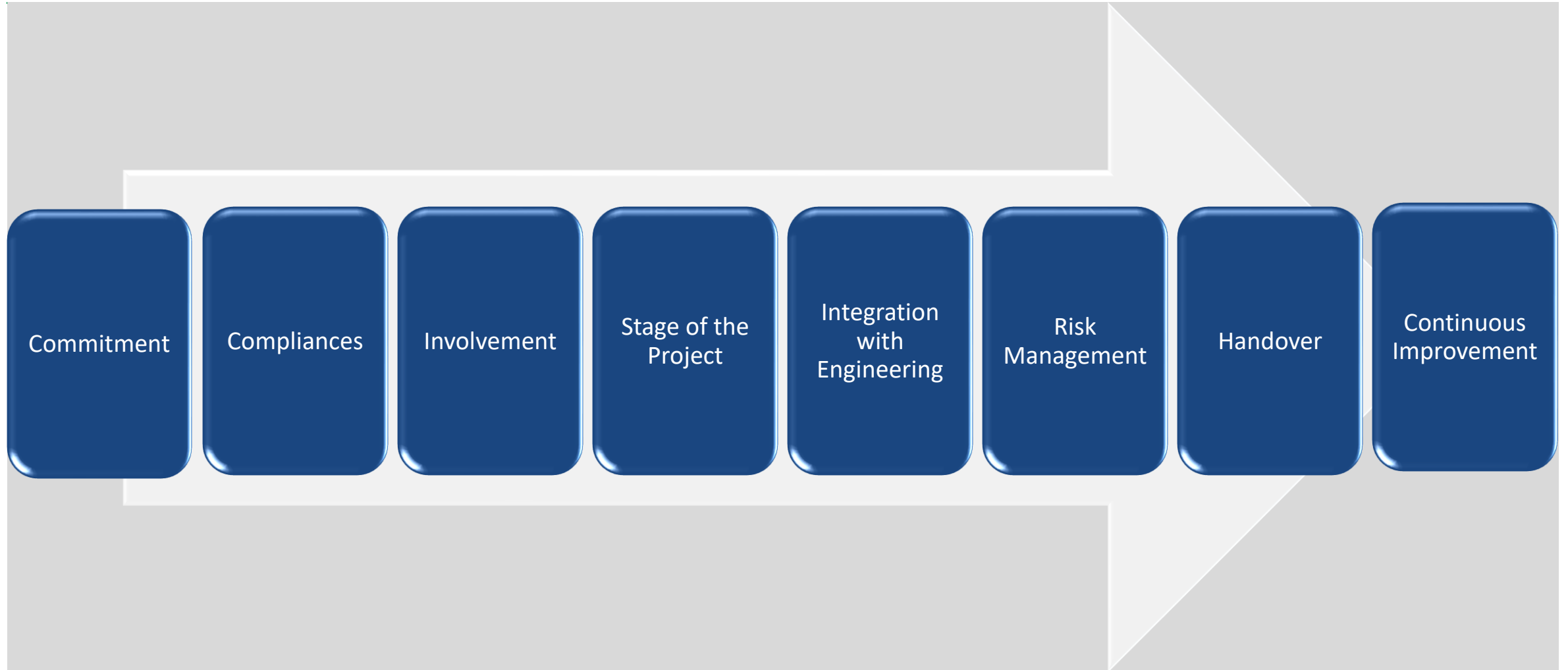


Continuous Improvement:

- ✓ Develop and follow incident investigation procedure
- ✓ Define and monitor key leading and lagging indicators
- ✓ Conduct internal and external audits and safety reviews
- ✓ Conduct motivational and welfare champions
- ✓ Act on the observations/recommendation ASAP
- ✓ Apply these lessons learned into new projects



Recap:

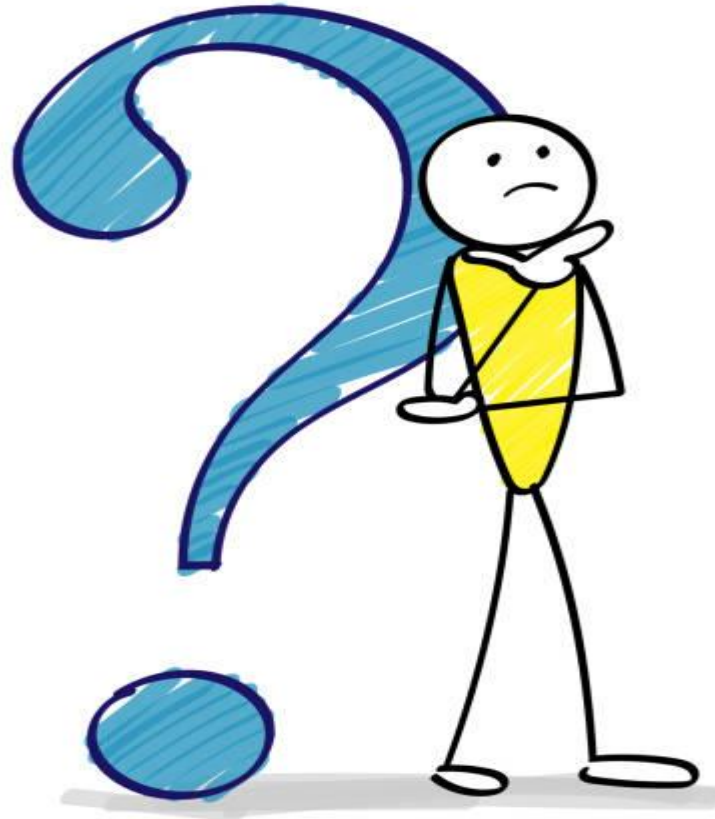


Conclusion:

- ✓ Management commitment is essential
- ✓ Early integration is key to success
- ✓ Process safety must be part of engineering design
- ✓ Safety-driven projects are sustainable projects
- ✓ Develop a procedure for Process safety integration

*“Accidents are not caused by a lack of knowledge,
but by a failure to use the knowledge,
that is available”.*

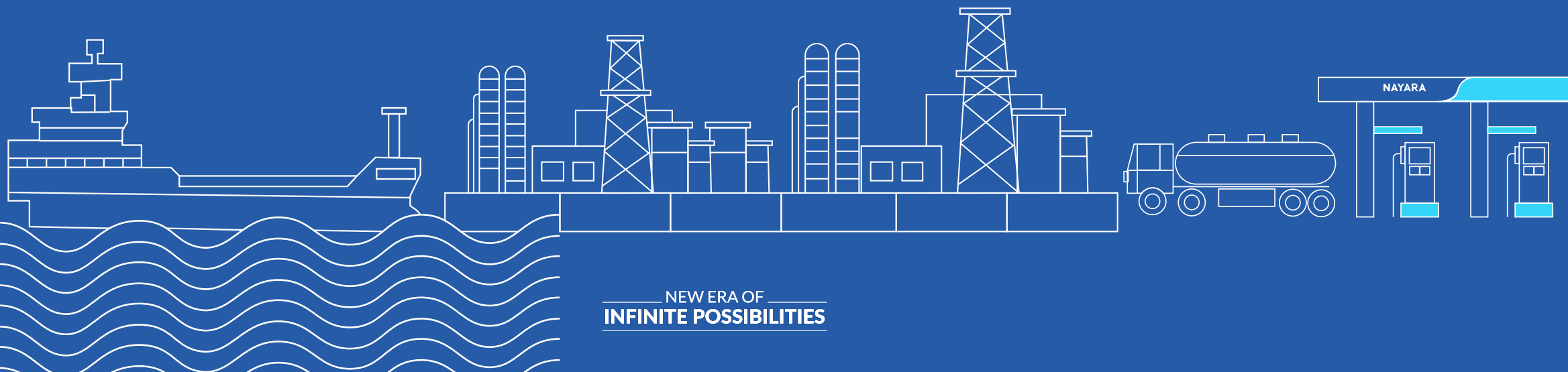
Dr. Travor Kletz



Source:

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INFINITE POSSIBILITIES

Thank You All



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